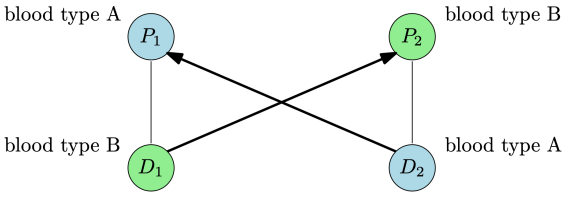
DS 574
Fall 2026

Kidney Exchange
Prof. Kira Goldner

Kidney Exchange

A patient–donor pair consists of a patient who needs a kidney and a willing donor who cannot donate to that patient because they are incompatible. In a two-way kidney exchange, two such pairs exchange donors: each donor gives to the other pair's patient, provided both donations are compatible.



Here P_i is patient i and D_i is their paired donor. The thin vertical lines identify the original pairs; the arrows show the cross donations $D_1 \to P_2$ and $D_2 \to P_1$.

Kidney exchange problem. Given patient–donor pairs and which donations are compatible, choose feasible exchanges. Each patient receives at most one kidney, and each donor gives at most one. We study exchanges without money, first using a housing-market analogy and then restricting to two-way exchanges to maximize the number of patients matched. How would one design a centralized mechanism for kidney exchange, where incompatible patient–donor pairs can register and be matched with others?

(a) **Who are the agents?** Patient–donor pairs, or hospitals, or both?

(b) **What's private?** What can they misreport, and how?

(c) **What's the objective?** Whose welfare, measured how?

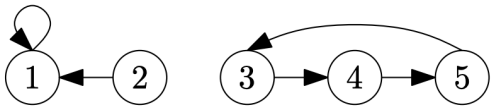
(d) **What's feasible?** Which exchanges can be carried out?

Idea #1: Top Trading Cycles

Consider the housing allocation problem defined by Shapley and Scarf [**?**]: there are n agents, and each initially owns one house. Each agent has a strict total ordering over the n houses, and need not prefer their own over the others. How can we reallocate the houses to make the agents better off?

Top Trading Cycles (TTC). While agents remain:

- Each remaining agent points to the owner of their favorite remaining house. This induces a directed graph on the remaining agents in which every vertex has out-degree 1.
- The graph has at least one directed cycle. Self-loops count.
- Execute every cycle: each agent gets the house owned by the agent they point to.
- Delete these agents and their houses, and repeat.



Find the cycles in this iteration. Who receives which house? What happens to agent 2? The cycles are (1) and $(3, 4, 5)$. Agents 1, 3, 4, 5 receive houses 1, 4, 5, 3 respectively. Agent 2 remains, and receives house 2 in the next iteration.

Every finite out-degree-one graph has a cycle: follow arrows until a vertex repeats. At least one agent leaves each iteration, and each departing agent receives exactly one house. An agent's own house remains available until they leave, so they weakly prefer their allocation to their endowment.

Theorem 1 (Stated without proof)**.** *With strict preferences, TTC is DSIC: for each agent, holding all other reports fixed, truthfully reporting their ranking is a dominant strategy.*

Individual rationality does not by itself prove DSIC. The strategic claim concerns a change in one report; the participation claim compares the allocation with the agent's own house.

Modifications for Kidney Exchange

The housing model is a useful starting point, but its assumptions matter. A TTC cycle can involve arbitrarily many agents. Here we restrict feasibility to pairwise exchanges, and model a patient as indifferent among compatible donors: utility is 1 if matched and 0 otherwise. These are modeling assumptions, not a claim that every kidney or every exchange program is identical. Patients without paired donors and donor-initiated chains require an extended model.

The lesson is to match the mechanism to the feasible exchanges and the information agents can hide. Coalition stability in a housing market does not establish truthfulness in a different reporting model.

Idea #2: Use a Matching Algorithm

The vertex set V consists of incompatible patient–donor pairs. An undirected edge $\{i, j\}$ means that donor i is compatible with patient j *and* donor j is compatible with patient i. A matching is a set of edges with no shared endpoints. Each edge implements two transplants, so maximizing the number of edges also maximizes the number of matched patients in this pairwise model.

Each agent i has a true set E_i of incident exchange edges and may report any subset $F_i \subseteq E_i$, but cannot invent a feasible exchange. The vertices and their priority order are fixed. The reported graph includes an edge only when both endpoints report it. Utility depends only on whether i is matched, not on which compatible donor they receive.

Priority Tie-Breaking

Fix the vertex order $1, 2, \ldots, n$ before reports arrive. First maximize matching cardinality; among those matchings, prefer matching vertex 1, then vertex 2 subject to the first decision, and so on.

1. Let $\mathcal{M}_0$ be the set of maximum-cardinality matchings of the reported graph.
2. For $i = 1, \ldots, n$, let Z_i be the matchings in $\mathcal{M}_{i-1}$ that match i. If Z_i is nonempty, set $\mathcal{M}_i = Z_i$; otherwise set $\mathcal{M}_i = \mathcal{M}_{i-1}$.
3. Return any matching in $\mathcal{M}_n$.

Each family remains nonempty. All final matchings cover the same vertices, so the remaining choice of edges is irrelevant to utilities.

Example. On the path $1 - -2 - -3 - -4 - -5$, list the maximum matchings and apply priority $1, 2, 3, 4, 5$. The maximum matchings are $\{12, 34\}$, $\{12, 45\}$, and $\{23, 45\}$, where ij denotes edge $\{i, j\}$. All have two edges. Priorities for vertices 1 and 2 retain the first two; priority for 3 selects $\{12, 34\}$. Vertex 5 is unmatched. **Same vertices versus different vertices.** Two matchings covering the same vertices give every patient the same utility. If their covered sets differ, each patient covered only by one strictly prefers that matching. Thus maximizing total transplants does not remove the need to choose which patients are served.

Theorem 2. *The priority matching mechanism is DSIC in the edge-hiding model: for any fixed reports of the other agents, hiding some incident edges cannot change an agent from unmatched to matched.*

Extension: Why Hiding Edges Cannot Help

Proof. Fix all other reports. If i is already matched when reporting E_i, their utility is 1 and cannot improve. Otherwise let M be a selected matching when i reports E_i. Hiding edges incident to the unmatched vertex i leaves every edge of M available. The new feasible matchings are a subset of the old ones, and M remains feasible.

The rule maximizes, in order, cardinality and then the matched/unmatched indicators of vertices $1, \ldots, n$. No remaining matching can beat M in this fixed order, since none beat it before. Thus every selected matching still has the same matched-vertex indicators as M, including an unmatched i. Hiding edges cannot help. □

This proof uses binary utilities, incident-edge deletion, a known vertex set, and priorities fixed independently of reports. It does not cover a hospital hiding several vertices.

Extension: Coalitions and the Core

Core: which agents, which deviation? Return to the housing model: every agent initially owns one house and has strict preferences over houses. An allocation assigns each house to exactly one agent. A coalition is a nonempty set of agents. It can leave the proposed allocation and redistribute only its members' *originally owned* houses, not the houses assigned to them by that allocation.

Strong core. A coalition weakly blocks an allocation if it can redistribute those original houses so that every member weakly prefers their new house to their house in the proposed allocation, and at least one member strictly prefers theirs. An allocation is in the *strong core* if no such coalition exists. Some conventions use "core" to rule out only deviations making every member strictly better off. We use the strong-core convention for the uniqueness theorem below.

Theorem 3. *With strict preferences, TTC returns the unique strong-core allocation.*

Proof. Let N_j be the agents removed in iteration j. For a proposed blocking coalition S, look at its members in the earliest TTC iteration that intersects S. Their assigned houses are their favorite houses among all houses then available; all endowments of S are still available. To be weakly better off, strict preferences force each of these members to keep their TTC house. The owner of each such house must also belong to S. Following the cycle forces the whole TTC cycle into S, with all its members receiving exactly their TTC houses.

Remove these cycles and repeat on the remaining members of S. Each member must receive exactly their TTC house, so no member improves strictly. The allocation is in the strong core.

For uniqueness, compare any other allocation with TTC and take the earliest iteration in which they differ. Earlier agents receive the same houses, so these houses are unavailable to later agents. In a cycle that differs, every member weakly prefers their TTC house to their house in the other allocation, and at least one strictly prefers it. The cycle's members

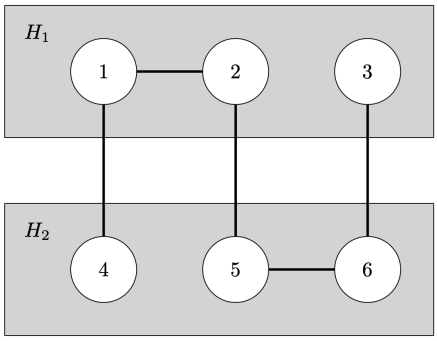
own the houses needed to execute it, so this coalition blocks the other allocation. Hence no different allocation is in the strong core. □

This is a coalition reallocation argument, not a proof that misreports are unprofitable.

Hospital Incentives

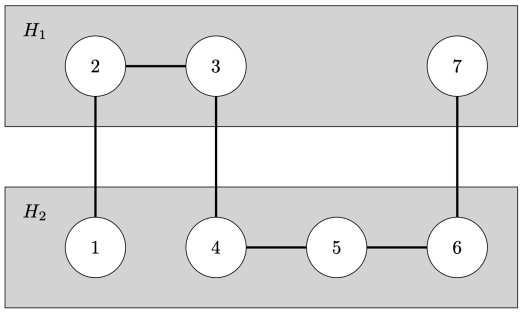
Hospitals may control which patient–donor pairs enter the exchange. A hospital's utility is the number of its own patients matched, including internal exchanges among pairs it withholds. Society counts all matched patients. This changes both the agent and the deviation: a hospital can hide vertices, not merely edges incident to one vertex.

The Need for Full Reporting



Compare internal matching first with reporting every pair. Matching internally gives $\{12, 56\}$ and matches four patients; the remaining vertices 3 and 4 have no edge. Full reporting permits $\{14, 25, 36\}$ and matches all six. This shows a loss from withholding internally matchable pairs, but not yet a strict gain for a hospital from withholding.

Hiding Patients



The graph is the seven-vertex path $1 - -2 - -3 - -4 - -5 - -6 - -7$. Hospital H_1 owns $\{2, 3, 7\}$ and H_2 owns $\{1, 4, 5, 6\}$. A maximum matching has three edges, leaving one patient unmatched.

If H_1 withholds 2 and 3, it matches them internally; the reported graph has the unique maximum matching $\{45, 67\}$. Thus all three of H_1's patients are matched. If H_2 withholds 5 and 6, it matches them internally; the reported graph has the unique maximum matching $\{12, 34\}$. Thus all four of H_2's patients are matched. Which hospital has a strict incentive depends on the full-report outcome.

Individual-pair DSIC therefore does not establish hospital DSIC. Approximation and hospital participation incentives are distinct goals in multi-hospital exchange [**?** **?**].

Extension: Hospital Impossibility

Theorem 4. *In the model above, no deterministic mechanism that always maximizes the number of matched pairs in the reported graph is DSIC for hospitals.*

Proof. Under full reporting, a maximum matching leaves exactly one of the seven patients unmatched. If that patient belongs to H_1, its full-report utility is 2; withholding 2 and 3 and using the internal exchange raises it to 3. If the unmatched patient belongs to H_2, its utility is 3; withholding 5 and 6 raises it to 4. Each reported graph has the unique maximum matching calculated above, so tie-breaking cannot avoid the profitable deviation. Some hospital can always gain. □

The impossibility requires the mechanism to maximize on every reported graph and allows withheld pairs to trade internally. It is not a claim about every conceivable hospital mechanism.

Extension: Two-Way versus Three-Way Exchange

Return to the feasibility assumption. Use a directed compatibility graph with an arc $i \to j$ when donor i can donate to patient j. Consider three pairs with precisely the arcs $1 \to 2$, $2 \to 3$, $3 \to 1$ and no reverse arcs.

The undirected pairwise-exchange graph has no edges, since no pair is mutually compatible. It matches no patients. The directed three-cycle matches all three. This isolates what the pairwise optimum does and does not optimize: maximum cardinality within one feasible set need not be optimal in an expanded feasible set. A directed compatibility arc here is not a TTC preference arrow.

Acknowledgements

This lecture was developed in part using materials by Tim Roughgarden, and in particular, his book "Twenty Lectures on Algorithmic Game Theory" [**?**].